

Will Wands

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Education

Rutgers University, Honors School of Engineering

New Brunswick, NJ

B.S. Electrical and Computer Engineering and Computer Science, GPA: 3.74

Expected May 2029

- Relevant Coursework: Data Structures, Discrete Mathematics, Data Management, Linear Algebra, Probability

Experience

The MITRE Corporation

Bedford, MA

Software Engineering Intern

May 2026 – Aug 2026

- Reduced translation latency by **30%** using **Python REST APIs** with **Redis** priority queues.
- Eliminated manual transfers between high-compliance systems using a **BlackSheep** API and **Vue.js** stack.
- Shipped production-ready code, resolving **10+ Jira tickets** on a 12-engineer **Agile** team using **Docker**, **Git**, **GitLab CI/CD**, and **Pytest** for unit and infrastructure testing.

NJ Governor's School of Engineering and Technology

New Brunswick, NJ

Research Scholar

Jul 2024 – Aug 2024

- Benchmarked **A*** and **RRT*** on metrics like path cost and runtime by deploying **Python** and **Pygame** simulated movement to a **Tello drone**.
- Published in **IEEE Xplore** after presenting at the **MIT Undergraduate Research Technology Conference**.

Projects

Imposture, Smart Posture Tracking Pet — 1st Place, ElevenLabs Track at YHack

Mar 2026

Python, OpenCV, MediaPipe, Arduino, C++, ElevenLabs API

- Detected real-time slouching using **Python/OpenCV** to score **MediaPipe** angles against calculated thresholds.
- Delivered voice, LCD, and haptic feedback to a 3D printed pet via **ElevenLabs API** and **C++** for an **Arduino**.

Teamfight Tactics Teacher, Gaming Stats Tracker

May 2026

C++, Python, REST API, Redis

- Computed 50+ match statistics with a **C++** engine and **JSON** output to a real-time **Python** dashboard.
- Limited Riot Games **API** requests using a custom **C++** key-value store with hash indexing and TTL eviction to avoid hitting the rate limit.

RUOnCampus, Off-Campus Housing Marketplace

Jan 2026 – May 2026

React, Express.js, PostgreSQL, Supabase, Tailwind CSS

- Cut search latency to under **40 ms** by normalizing **8 PostgreSQL tables** and indexing high-volume queries.
- Reduced onboarding drop-off by **25%** with a **React** flow, debounced **Express.js** lookups, and JWT authentication.

Angry Signing Llama, Real-Time Computer Vision Based ASL Pipeline

Feb 2026

Python, OpenCV, FastAPI, TensorFlow, MediaPipe, JavaScript, Keras

- Achieved **97%** accuracy across **35** static and dynamic signs using **TensorFlow Convolutional Neural Network (CNN)** and **Recurrent Neural Network (RNN)** models trained on **MediaPipe** keypoints.
- Delivered sub-second inference over **WebSockets** between a **Python/OpenCV** pipeline and **JavaScript** frontend.

Viand, Group Dining Assistant

Jul 2026

Next.js, TypeScript, Claude API, React, Telegram Bot API

- Built a **Next.js** and **React TypeScript** dining messaging service for iMessage and **Telegram** groups using **OpenStreetMap's Overpass API** and **Claude APIs** within a 5-mile radius.

Leadership

IEEE Intelligent Ground Vehicle Competition

Rutgers University

Lead Programmer

Sep 2025 – Present

- Implemented a **ROS 2** obstacle avoidance stack by using **LiDAR** and an **OAK-D camera** for localized mapping.
- Simulated the vehicle and course in **Gazebo** using **XML/SDF** models on **Linux** to allow efficient testing.

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, SQL, HTML, CSS

Frameworks and Libraries: FastAPI, Next.js, Express.js, React, Vue.js, Tailwind CSS, BlackSheep, ROS 2, TensorFlow, Keras, OpenCV, MediaPipe

Infrastructure and Tools: Linux, Git, Docker, Kubernetes, AWS, Redis, PostgreSQL, Supabase, CI/CD

Certifications and Programs: AWS Certified Cloud Practitioner, AI4ALL Scholar